HARSHVARDHAN BIRLA

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EXPERIENCE

Oracle Bengaluru, India

Software Engineer, Database Security Team *Jan 2025 – Present*

- Resolved **100+** customer-reported and internally detected production defects in C-based Oracle RDBMS infrastructure, including memory corruption, null-pointer crashes, stack-buffer overflows, ORA-07445/ORA-00600 failures, and reliability regressions using GDB, core dumps, trace files, and internal diagnostic tooling.
- Designed and implemented an enhancement to **Oracle Label Security (OLS)**, expanding supported label capacity by **8x** from `VARCHAR2(4000)` to `VARCHAR2(32000)` to support larger organizations requiring security labels with more groups and compartments while preserving database-level access-control behavior.
- Fixed **40+** security-critical defects across multiple RDBMS subsystems, including heap overwrites, format-string vulnerabilities, uninitialized-memory defects, and SQL/LDAP injection exposure.
- Owned engineering and investigation work across Oracle Label Security and password-file components, covering access-control flows, authentication, role systems, session state, and client–database consistency mechanisms.
- Drove cross-version feature integration from Oracle 26ai/23c to 19c, adapting low-level database internals while maintaining backward compatibility and consistent behavior across database releases.

iQuadra Information Services Pvt. Ltd Remote

Machine Learning Intern *May 2023 – Jul 2023*

- Fine-tuned a RoBERTa-based sentiment model and developed computer-vision pipelines for candidate assessment, improving interview evaluation accuracy by **35%**.

PROJECTS

Distributed Storage Engine | ***Apache Kafka, Replication, Fault Tolerance*** Jul 2023 – Nov 2023

- Architected a multi-node distributed storage system using Apache Kafka for scalable ingestion, partitioned inter-node communication, and durable data movement.
- Implemented replicated metadata and dual-storage fault-tolerance mechanisms, enabling continued data access and recovery during tested single-node failure scenarios.
- Optimized inter-node communication and concurrent processing through Kafka topic partitioning and parallel consumers, reducing transfer latency and improving throughput.

Live Snapshot Prune | ***Python, PostgreSQL, Docker, Distributed Storage*** 2026

- Designed a backup storage system with transactional PostgreSQL metadata, independent object storage, snapshot APIs, and online mark-and-sweep garbage collection running alongside concurrent uploads, restores, and deletions.
- Designed correctness mechanisms that distinguish logical content identity from physical object incarnations, preventing delayed delete operations from removing newly uploaded deduplicated data during ABA-style races.
- Built a fault-injection verifier covering lost acknowledgements, process crashes, restart recovery, concurrent operations, orphan reclamation, and end-state invariant validation across multiple services.

Low-Latency C++ Event Processing Engine | ***C++17, Multithreading, Lock-Free Queue, SQLite*** Jul 2025

- Built a modular C++17 event-processing backend with live and replayable data sources, circuit-breaker-based provider fault handling, explainable alerts, and SQLite-backed persistence paths.
- Implemented lock-free inter-thread event handoff and benchmarked the in-memory processing path at **8.37M events/sec** with **10.9μs p99** local decision latency.
- Developed a reproducible validation workflow covering unit tests, deterministic replay, threshold-sweep backtesting, latency benchmarks, labeled failure scenarios, and circuit-breaker state transitions.

Distributed Compute Orchestration Platform | ***HDFS, Redis, ZooKeeper*** Dec 2023 – Apr 2024

- Engineered distributed execution infrastructure for scalable multi-node task orchestration with fault-tolerant coordination and distributed state management.
- Integrated HDFS for durable storage, Redis for low-latency state and cache access, and ZooKeeper for distributed coordination across worker nodes.
- Designed optimized data pipelines to reduce inter-component latency and improve the reliability of distributed workflow execution.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, SQL, PL/SQL
Data & Infrastructure: Oracle RDBMS, PostgreSQL, Apache Kafka, ZooKeeper, Redis, HDFS, SQLite
Systems: Distributed Systems, Storage Systems, Database Systems, Concurrency, Multithreading, Fault Tolerance, High Availability, Low-Latency Systems, Data Structures & Algorithms
Tools: Linux, GDB, Docker, Git, CMake, Gradle, Shell Scripting, REST

EDUCATION

BITS Pilani, Goa Campus Goa, India

B.E. (Hons.) in Computer Science, ***CGPA: 8.7*** *2021 – 2025*

- Relevant Coursework: Operating Systems, Distributed Systems, Database Systems, Computer Networks, Object-Oriented Programming, and Design and Analysis of Algorithms

ACHIEVEMENTS

NTSE Scholar: Awarded to the **Top 1000** students across India | **JEE Mains:** 99.69 percentile | **JEE Advanced:** AIR 3547